

WASP-11b

R-1043

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_161020 · created 2026-07-30T08:30:41+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457681.8364 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.109 +/- 0.052
Transit depth (Rp/Rs)^2	1.18 +/- 1.13 [%]
Orbital Inclination (inc)	88.25 +/- 1.73
Airmass coefficient 1 (a1)	0.31 +/- 0.21
Airmass coefficient 2 (a2)	1.08 +/- 1.28
Scatter in the residuals of the lightcurve fit is	65.09 %
Best Comparison Star	#3 - [287, 384]
Optimal Aperture	0.99
Optimal Annulus	4.75
Transit Duration (day)	0.091 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.109 $\pm$ 0.052 vs 0.131 (deviation 0.42 σ)
Duration fit vs published	0.091 d vs 0.1075 d (deviation 0.75 σ)
Mid-time O–C	4.1 min
Depth SNR	2.1
Residual scatter	65.09 %

Light curve

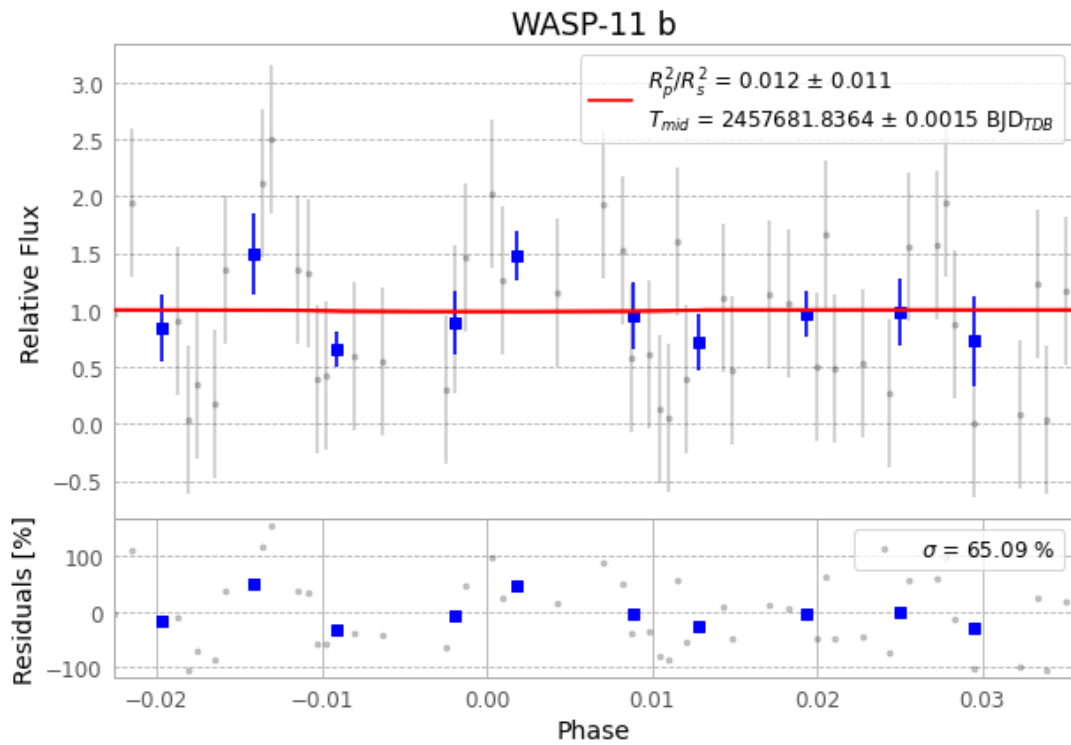


Figure 1 — FinalLightCurve_WASP-11 b_2016-10-19.png (SHA-256 72d8eb8eddd7a196...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [580, 325], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ca2a882d1e25f907...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

107 FITS frames (70 MB), HATP-10161020054812.FITS → HATP-10161020110612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-161020.log (46e0dfd2fbeb...), FinalLightCurve_WASP-11 b_2016-10-19.png (72d8eb8eddd7...), FinalLightCurve_WASP-11 b_2016-10-19.csv (ff2147f10b3f...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 08:30Z.